MST. FAHMIDA AKTER

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RESEARCH INTEREST

AI & ML , AI Agents, Biomedical Image Analysis, Computational Biology, Human-Centered AI

EDUCATION

Varendra University Rajshahi, Bangladesh
B.Sc. in Computer Science and Engineering (CGPA: 3.86/4.00) *Summer 2022 – Spring 2026*

ACHIEVEMENTS

- **NASA Space Apps Challenge 2023 (World Champion):** Team Voyagers - Won the "Best Storytelling" award worldwide for "Aqua Explorer", a solution addressing the "Everything Starts with Water" challenge, recognized for outstanding innovation and presentation.
- **Hult Prize 2023:** New York Summit Round Participant (virtual) and On-Campus Champion (Team Salvage).
- **International Blockchain Olympiad 2024 (Finalist):** Nominated by BCOLBD and selected as a finalist in the Netherlands round.
- **Undergraduate Conference on Intelligent Computing and Systems 2026 (UCICS):** 2nd Runner-Up in Poster Presentation.
- **100% Merit-Based Scholarship** - Awarded full-tuition scholarship for the B.Sc. program based on academic result

RESEARCH PUBLICATION

Munam, A. M., & **Akter, M. F.** (2026). **Parent-daughter relationship quality and depressive symptoms among unmarried adolescent girls in Bangladesh: Evidence from 2019-20 Bangladesh Adolescent Health and Wellbeing Survey.** *Psychiatry Research Communications*, 6(1), 100244. `https://doi.org/10.1016/j.psycom.2026.100244`

SUBMITTED TO JOURNALS

Baghchesaraei, O. R., Choudhary, R., Paneerselvam, B., Duraisamy, Y., Gupta, R., Bhagat, C., Ramu, P., **Akter, M. F.**, & Enefu, O. M. **Chapter 9: AI for Waste Management, Circular Economy, and Sustainability Optimisation.** In: *Artificial Intelligence in Environmental Science: Opportunities, Challenges, and Ethical Frameworks*. Bentham Science Publishers.

UNDER EDITING / WORK IN PROGRESS

- **HemoCount: A Web-Deployable System for Automated Hemocytometer Cell Counting Using a Lightweight YOLO Model**
  Journal manuscript under preparation and editing.

RESEARCH EXPERIENCE

Undergraduate Research 2025 – 2026
Supervisor: Prof. Shamim Ahmad (Head, Dept. of CSE, Varendra University)

Conducted research on automated cell counting, developing a lightweight YOLO model for hemocytometer cell counting.

WORK EXPERIENCE

Project Lead – Youth Leadership Training 2023 2023
Social Saviors Project

Led a funded project, "Social Saviors," to raise awareness among students about the harmful effects of excessive Facebook usage.

PROJECTS

Bookie | *Flutter, Firebase, Android Studio* Software Development Lab II

– A cross-platform mobile app for book lovers featuring book discovery, reviews, and personalized recommendations.

Scholarly Way | *HTML, CSS, JavaScript, Python Django, SQLite3* Software Development Lab III

– A web-based platform that helps students find global scholarships using country-specific and ranking-based filters.

IoT Air Quality Monitor | *ESP32, Blynk IoT, C++* Microcontroller Lab

– Engineered a real-time environmental monitor using ESP32 integrated with gas, temperature, dust, and rain sensors.

– Built a custom Blynk IoT dashboard for live cloud-based data visualization and remote tracking.

– Implemented an automated LED alert system, applying core IoT and embedded systems design principles.

EXTRACURRICULAR ACTIVITIES

– **Campus Director** of **Hult Prize** at **Varendra University** (2025-26)

– **Co-Convenor 1 - BASIS Forum, Varendra University Chapter** (2024-25)

– **Campus Ambassador - Excellence Bangladesh, Varendra University Team** (2026)

– **She Squad Leader - BDapps** (2023-2025)

– **Judges Support Head & Expert - Hult Prize Varendra University** (2023-2024)

– **Completed Youth Leadership Training** - 2022

SKILLS

Programming Languages: C, C++, R, Python
Data Analysis & Databases: Power BI, Tableau, Power Query, Excel, MySQL, Data Manipulation, Python for Data Analysis, SQL
Tools & Applications: GitHub, Figma, Canva, Roboflow, Zotero, LaTeX, Microsoft Word
Soft Skills: Leadership, Communication, Problem Solving, Time Management, Adaptability, Team Collaboration